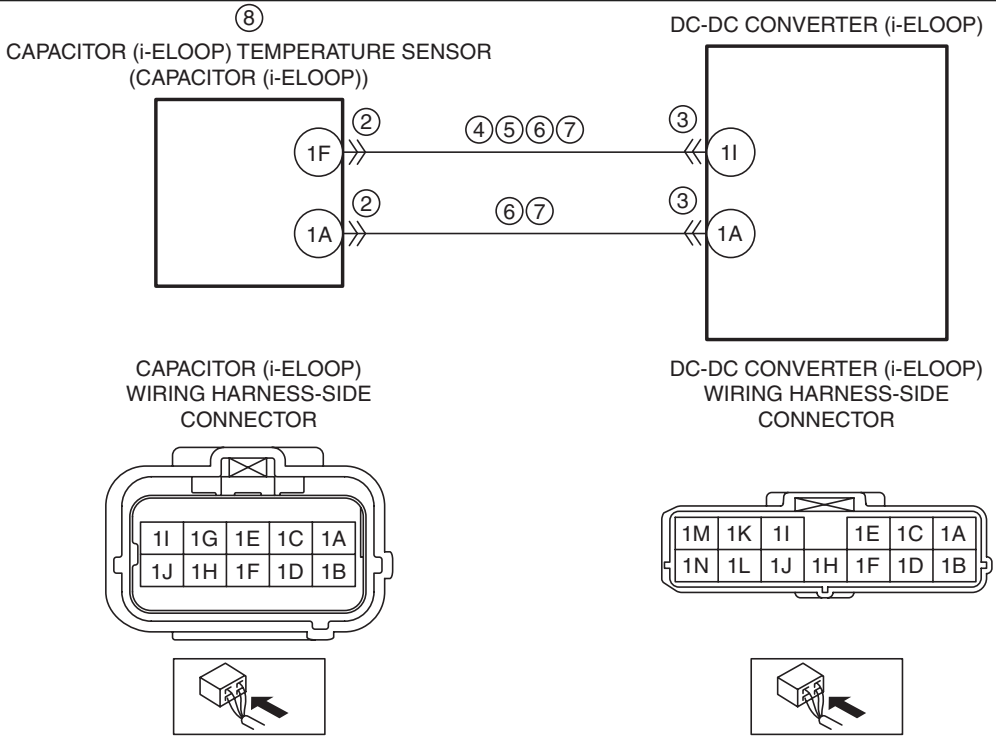
DTC P0C39:00 [i-ELOOP]

id130201000800

Note
- If the capacitor (i-ELOOP) voltage is low after replacing the DC-DC converter (i-ELOOP), the i-ELOOP indicator light (green) flashes until the capacitor (i-ELOOP) voltage is **approx. 9.5 V**.

DTC P0C39:00	**Capacitor (i-ELOOP) temperature sensor system: function/circuit malfunction**
DETECTION CONDITION	• Any one of the following conditions is met: — A condition in which the capacitor (i-ELOOP) temperature sensor voltage is **less than 0.035 V** (equal to output value at capacitor (i-ELOOP) temperature of **139 °C {282 °F} or more**) continues for **5 s**. — A condition in which the capacitor (i-ELOOP) temperature sensor voltage is **3.46 V or more** (equal to output value at capacitor (i-ELOOP) temperature of **−70 °C {−94 °F} or less**) continues for **5 s**.
FAIL-SAFE FUNCTION	• Not applicable
POSSIBLE CAUSE	• Capacitor (i-ELOOP) connector or terminals malfunction • DC-DC converter (i-ELOOP) connector or terminals malfunction • Short to ground in wiring harness between capacitor (i-ELOOP) terminal 1F and DC-DC converter (i-ELOOP) terminal 1I • Short to power supply in wiring harness between capacitor (i-ELOOP) terminal 1F and DC-DC converter (i-ELOOP) terminal 1I • Capacitor (i-ELOOP) temperature sensor signal circuit and ground circuit are shorted to each other • Open circuit in wiring harness between the following terminals: — Capacitor (i-ELOOP) terminal 1F—DC-DC converter (i-ELOOP) terminal 1I — Capacitor (i-ELOOP) terminal 1A—DC-DC converter (i-ELOOP) terminal 1A • Capacitor (i-ELOOP) temperature sensor malfunction • DC-DC converter (i-ELOOP) malfunction



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
2	**INSPECT CAPACITOR (i-ELOOP) CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Disconnect the capacitor (i-ELOOP) connector. • Inspect for poor connection (such as damaged/pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
3	**INSPECT DC-DC CONVERTER (i-ELOOP) CONNECTOR CONDITION** • Disconnect the DC-DC converter (i-ELOOP) connector. • Inspect for poor connection (such as damaged/pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
4	**INSPECT CAPACITOR (i-ELOOP) TEMPERATURE SENSOR SIGNAL CIRCUIT FOR SHORT TO GROUND** • Verify that the capacitor (i-ELOOP) and DC-DC converter (i-ELOOP) connectors are disconnected. • Inspect for continuity between capacitor (i-ELOOP) terminal 1F (wiring harness-side) and body ground. • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between capacitor (i-ELOOP) terminal 1F and DC-DC converter (i-ELOOP) terminal 1I. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 9.
		No	Go to the next step.
5	**INSPECT CAPACITOR (i-ELOOP) TEMPERATURE SENSOR SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the capacitor (i-ELOOP) and DC-DC converter (i-ELOOP) connectors are disconnected. • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the DC-DC converter (i-ELOOP) detecting an open circuit. • Measure the voltage at the capacitor (i-ELOOP) terminal 1F (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between capacitor (i-ELOOP) terminal 1F and DC-DC converter (i-ELOOP) terminal 1I. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 9.

STEP	INSPECTION		ACTION
6	**INSPECT CAPACITOR (i-ELOOP) TEMPERATURE SENSOR SIGNAL CIRCUIT AND GROUND CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the capacitor (i-ELOOP) and DC-DC converter (i-ELOOP) connectors are disconnected. • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect for continuity between capacitor (i-ELOOP) terminals 1F and 1A (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Capacitor (i-ELOOP) terminal 1F—DC-DC converter (i-ELOOP) terminal 1I • Capacitor (i-ELOOP) terminal 1A—DC-DC converter (i-ELOOP) terminal 1A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 9.
		No	Go to the next step.
7	**INSPECT CAPACITOR (i-ELOOP) TEMPERATURE SENSOR CIRCUIT FOR OPEN CIRCUIT** • Verify that the capacitor (i-ELOOP) and DC-DC converter (i-ELOOP) connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side): — Capacitor (i-ELOOP) terminal 1F—DC-DC converter (i-ELOOP) terminal 1I — Capacitor (i-ELOOP) terminal 1A—DC-DC converter (i-ELOOP) terminal 1A • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Capacitor (i-ELOOP) terminal 1F—DC-DC converter (i-ELOOP) terminal 1I • Capacitor (i-ELOOP) terminal 1A—DC-DC converter (i-ELOOP) terminal 1A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
8	**INSPECT CAPACITOR (i-ELOOP) TEMPERATURE SENSOR** • Inspect the capacitor (i-ELOOP) temperature sensor. (See CAPACITOR (i-ELOOP) INSPECTION [i-ELOOP].) • Is there any malfunction?	Yes	Replace the capacitor (i-ELOOP), then go to the next step. (See CAPACITOR (i-ELOOP) REMOVAL/INSTALLATION [i-ELOOP].)
		No	Go to the next step.
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) • Switch the ignition off. • Switch the ignition ON (engine on) and wait for **1 s or more**. • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) — Start the engine and leave for **100 s or more**. — Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
10	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [i-ELOOP].)
		No	DTC troubleshooting completed.